COURSE SYLLABUS

PROGRAM	MSc & PhD in Administration	
COURSE	Panel Data	
TERM AND WORKLOAD	4th Quarter	30 h
PROFESSOR	Rodrigo Leite	
E-MAIL	rodrigo.oliveira@coppead.ufrj.br	
CLASS SCHEDULE	Tuesdays, from 2 PM to 5:20 PM	
COURSE PERIOD	October 20th to December 15th	

PROGRAM

COURSE DESCRIPTION

The aim of this course is to Introduce the distinctive features of panel data. Review some panel data sets commonly used in social sciences. Present the advantages (and limitations) of panel data, and consider what sort of questions panel data can(not) address. Also, to introduce the basic estimation techniques for panel data (linear and non-linear) and discuss how to choose (and test for) the right technique for the question being addressed. Finally, we discuss how to interpret the results and how to derive economic implications. Additionally, some techniques for time series estimations and emerging topics (AI and Bayesian) will be discussed.

Part of the classes will consist of hands-on work on Stata, so all students should have their computers equipped with Stata on hand.

Students are asked to answer the Panel Data Pre-Class Assessment before the first class via this link: https://forms.gle/v8q37nCiFd2wdoiX6

It is very important that you answer the survey above, in order to help me tailor the course to be as useful as possible for you.

Class Dates:

OCT: 27 | 30
NOV: 3 | 10 | 17 | 24
DEC: 1 | 8 | 15

SCOPE & OBJECTIVES

This course aims to equip MSc and PhD students with the conceptual and applied tools needed to design, estimate, and interpret panel data models for empirical research in business, economics, and public administration. By combining the core panel data toolkit with selected extensions (time series, Bayesian methods, and the use of AI in empirical research) the course lays a methodologically rigorous foundation that students can apply directly to their own dissertation or thesis work, across research lines.

By the end of the course, students should be able to:

- Understand the distinctive features of panel data and the kinds of research questions it can (and cannot) address, relative to purely cross-sectional or time-series data.
- Estimate and correctly choose among core panel data models, such as pooled OLS, fixed effects, and random effects, using the appropriate specification tests (Hausman, Breusch-Pagan).

- Apply these methods hands-on in Stata, gaining practical fluency in estimating, diagnosing, and validating panel data models on real datasets.
- Extend the core toolkit to more advanced techniques, including instrumental variables, differences-in-differences, dynamic panel models (GMM), time series models (ARIMA, GARCH), and Bayesian regressions.
- Critically read, present, and discuss published empirical papers that use panel data methods, evaluating their identification strategy and contribution.
- Design a well-motivated research proposal that applies panel data methods to an original question, building the foundation for a full academic paper.
- Engage critically with emerging tools, including the use of AI in empirical research and replication practices, as part of a rigorous research process.

METHODOLOGY

The course will consist on lectures and Panel Data implementations. Also, there will be Problems sets and Readings to be discussed in class.

All students attending the class are expected to read the papers selected for each topic in advance and to participate in class discussions. Auditors (if any) are expected to do all of the assignments of regular students, except the mid term and final evaluations.

For most PhD students, the most difficult part of the program is the search for a dissertation topic. To acquire some experience selecting research topics, a main course requirement is to prepare a proposal for a project concerning issues we discuss in class. I hope that a set of research ideas will come to you as the course progresses. This set will presumably consist of multiple ideas in different stages. While some may be little more than preliminary (but hopefully thought-provoking) idea stubs, at least one should be in a more fully-formed stage by the end of the quarter. For the latter, you should target writing down a document containing:

(1) One or two pages motivating the idea/research question. You want to identify a question, explain why it is important that we know more about it (i.e., its implications for policy making, etc.), and make the case that the existing literature has ignored or overlooked the issue.
(2) Once you motivate your research question, you need to think about the economics of your inquiry. This would probably take a page or two. For example, what are the tradeoffs involved? What kinds of corporate actions/consequences do they entail? Should we expect to verify proposed effects using real world data? What are the exact methods we should use and why?
(3) Anticipating the results: Based on priors, what results should be expected? (What if you find something else?) Discussing this should take a few paragraphs.
(4) Add a literature review. In this profession, it is important that you recognize (and give due credit to) those who did prior research in the area you want to do your first paper. Do a fair (not aggressive, but still critical) review of the literature in no more than three pages. The emphasis should be on establishing your case that the "void" in the literature is worth pursuing. Try to organize the papers in a compare/contrast format. Like all you do in the paper, use this to motivate the importance of your research question.
(5) Take a page or two to discuss "implementation issues" such as data availability, sample construction, programming needs, replicability, etc.
(6) Have some preliminary results in the manuscript. These can be empirical results, simulations/calibrations, or anything that gives me hope that this is more than a bunch of nice sounding paper ideas that will never see the light of day.

(7) A final, polished manuscript should be sent to me up to December 8th at noon.

Importantly, the proposed paper has to be about ideas related to the class. It is not acceptable for students to simply present a paper that was already ready in their pipeline. The paper has to use the methods discussed in class.

ASSESSMENT PROCEDURES

- Paper presentation: 10%

Presentation of papers for each class. The number of presentations for each student will be decided during the class, depending on how many will attend.

- Assignments: 30%

I will give specific guidelines regarding homework assignments. One will be data-oriented, hands-on, which will be delivered on Class 8. The second will consist of you writing a review on a real paper, I will provide you with the paper and you will have two weeks to write the review.

- Midterm evaluation: 30%

It will consist of a writing exam.

- Research Project presentation: 30%

You will have 15-20 minutes to present the paper in the last class, and then we will have about 5-10 minutes to discuss the paper. Usually at some conferences you have a dedicated discussant that has read the paper and will give you feedback on the topics. In this "mock-up conference" I will be the discussant for each paper; hence that's why you must send me the papers by December 8th, otherwise I will not be able to read your proposals and give you proper feedback.

DETAILED COURSE CONTENT

Class 1: The Basics of Panel Data
Hour 1: Introduction and Why Study Panel Data?
Hour 2: Panel Data Structure
Hour 3: Pooled OLS

Class 2: Panel Data Regressions
Hour 1: Random vs Fixed Effects
Hour 2: Hausmann and Breusch-Pagan Tests
Hour 3: Paper Presentations and Discussions

Class 3: Advanced Panel Data Models I
Hour 1: Pseudo-Panels
Hour 2: IV Regressions in Panel Models
Hour 3: Paper Presentations and Discussions

Class 4: Advanced Panel Data Models II
Hour 1: Differences-in-Differences in Panel Models
Hour 2: Dynamic Models: Autocorrelation and GMM
Hour 3: Paper Presentations and Discussions

Class 5: Midterm Exam (Hours 1-3)

Class 6: Hands-On
Hour 1: Discussion on the Midterm
Hour 2: Panel Data Exercise on Stata

Hour 3: Paper Presentations and Discussions

Class 7: Panel Data Extensions I: Time Series
Hour 1: AR, MA, ARMA, and ARIMA models
Hour 2: ARCH and GARCH models
Hour 3: Paper Presentations and Discussions

Class 8: Panel Data Extensions II: Bayesian Models
Hour 1: Introduction to Bayesian Models
Hour 2: Bayesian Regressions
Hour 3: Paper Presentations and Discussions

Class 9: AI and Mockup Conference
Hour 1: The Use of AI in Academic Research
Hour 2 & 3: Mock-up Conference

Papers to read and present each class:

Class 2:

- R. Leite, R. Cardoso, A. Jelihovschi and J. Civitarese (2020) Job market compensation for cognitive reflection ability. *Research in Economics*, 74(1): 87-93.
- L. Moura and L. Norden (2019) Does good corporate governance pay off in the long run? Evidence from stock market segment switches in Brazil. *Brazilian Review of Finance*, 17(3): 1-25.
- E. Fama and K. French (2015) A five-factor asset pricing model. *Journal of Financial Economics*, 116(1): 1-22.

Class 3:

- R. Leite, L. Mendes and R. Moreira (2020) Profit Status of Microfinance Institutions and Incentives for Earnings Management. *Research in International Business and Finance*, 54: 101255.
- L. Pilla, A. Peci, & R. Leite. Financial Performance of State-Owned Enterprises: Does Political Ideology Play a Role? *Journal of Public Administration Research and Theory* 35(3): 338–351
- W. Jiang (2017) Have Instrumental Variables Brought Us Closer to the Truth? The Review of Corporate Finance Studies, 6(2): 127–140

Class 4:

- R. Leite, L. Mendes and L. Sacramento (2019) To Profit or not to Profit? Assessing financial sustainability outcomes of Microfinance Institutions. *International Journal of Finance & Economics*, 24(3): 1287-1299.
- L. Pilla, A. Peci, & R. Leite. (2026) State-Owned Enterprises, Public Corporate Governance Codes, and Financial Performance: Evidence from Brazil. *Corporate Governance: The International Journal of Business in Society*, forthcoming.
- L. Fang, J. Goldman, & A. Roulet. (2026) Private Equity and Pay Gaps Inside the Firm. *Journal of Finance*, forthcoming

Class 6:

- L. Pilla, A. Peci, & R. Leite. Is the State a Socially Responsible Shareholder? State-Owned Enterprises, Political Ideology, and Corporate Social Performance. *Journal of Business Ethics*, 197(2): 289-309.

- R. Leite, L. Mendes, & I. Coutinho. The Political Economy of Banks' Cash Holdings. *Journal of Economic Studies*, 52(8): 1503–1518.
- A. Berger, C. Ortega, M. Busch, & R. Roman. (2026). Banking on deforestation: The cost of nonenforcement. *Journal of Financial Intermediation*, 67: 101211.

Class 7:

- L. Mendes, R. Leite, & J. Fajardo. (2022). Do Contingent Convertible Bonds Reduce Systemic Risk? *Journal of International Financial Markets, Institutions and Money*, 78: 101554.
- R. Leite, L. Mendes, & P. Soares. (2026). Who Benefits From Government-Backed Debt Renegotiation? Evidence From a Natural Experiment in Brazil. *SSRN Working Paper*.
- A. Menkveld, et al. (2024). Nonstandard errors. *Journal of Finance*, 79(3): 2339-2390.

Class 8:

- R. Leite (2025). The Public Cost of Economic Policy Uncertainty: Evidence from Brazilian Public Procurements. *Finance Research Letters*, 86(D): 108665.
- R. Leite, L. Mendes, & I. Coutinho (2025). The Political Economy of Banks' Cash Holdings. *Journal of Economic Studies*, 52(8): 1503–1518.
- K. Lang, K. Leong, H. Li, & H. Xu. (2025). Borrowing in an illegal market: Contracting with loan sharks. *Review of Economics and Statistics*, 107(1), 269-278.

BIBLIOGRAPHY	
Required Reading	See syllabus above.
Supplementary reading	Arellano, M. (2003), Panel Data Econometrics (Advanced Texts in Econometrics), Oxford University Press. Bond, S.R. (2002): "Dynamic Panel Data Models: A Guide to Micro Data Methods and Practice". Cemmap Working Paper Series No. CWP09/02, Institute for Fiscal Studies, London. Greene, W. (2000): Econometric Analysis (5th Edition), Prentice Hall Roodman, D. (2006): "How to Do xtabond2: An Introduction to "Difference" and "System" GMM in Stata". Working Paper 13, Centre for Global Development. Wooldridge, J. M. (2001), Econometric Analysis of Cross Section and Panel Data, MIT Press. Cameron and Trivendi (2010), Microeconometrics Using Stata. STATA Press
Cases / Activity	See syllabus above.
Pre/ Post Assignment Deadline	See syllabus above.

LEARNING OBJECTIVES AND IMPACT ON THE ENVIRONMENTAL, SOCIAL AND GOVERNANCE AGENDA

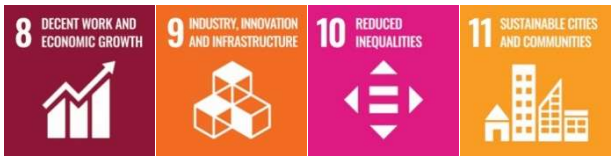
LEARNING OBJECTIVES

Competency:	Course objectives:	Level of contribution *(Click to select):*
Conceptual Skills	Understand the theoretical foundations of panel data models (pooled OLS, fixed effects, random effects, dynamic panels), their identifying assumptions, and when each is appropriate for a given research design.	☐☐☒
Methodological Skills	Develop applied competence in estimating and diagnosing panel data models using statistical software (Stata), including specification tests (Hausman, serial correlation, heteroskedasticity), robustness checks, and basic treatment of endogeneity.	☐☐☒
Communication Skills	Build the ability to present and interpret panel data results clearly in seminars and in written empirical work, including justifying model choice to an academic audience.	☐☒☐
Dissemination Skills	Prepare students to translate panel data analyses into the methodology and results sections of academic papers suitable for peer-reviewed submission.	☐☒☐
Relevance	Provide a methodological toolkit directly applicable to students' own dissertation/thesis research involving longitudinal, cross-sectional, or time-series data.	☐☐☒

IMPACT ON THE ENVIRONMENTAL, SOCIAL, AND GOVERNANCE AGENDA

Upon completing this course, the student will be able to draw connections between the content covered and the potential environmental, social, and governance impact, as represented by the following Sustainable Development Goals (SDGs):

(Maintain discipline-related SDGs)



ABOUT THE PROFESSOR

	PROFESSOR'S NAME: Prof. Dr. Rodrigo Leite



SHORT CV:

Prof. Rodrigo Leite holds a PhD in Business from EBAPE/FGV and is Assistant Professor of Finance at COPPEAD/UFRJ. He was also a Visiting Professor at INSEAD (France) and CUNEF Universidad (Spain) as well as a former Assistant Professor of Accounting at Rio de Janeiro State University. His research has been published in several finance, economics, accounting, and administration journals such as the "Journal of Public Administration Research and Theory" (ABS 4), "Journal of Business Ethics" (ABS 3), "The International Journal of Accounting" (ABS 3), "Accounting, Auditing and Accountability Journal" (ABS 3), "Accounting Forum" (ABS 3), "Journal of International Financial Markets, Institutions and Money" (ABS 3), "International Journal of Finance and Economics" (ABS 3), "Finance Research Letters " (ABS 2), "Journal of Economics and Business" (ABS 2), "Research in International Business and Finance" (ABS 2), "International Review of Economics and Finance" (ABS 2), "Journal of Economic Studies" (ABS 2), "Journal of Behavioral and Experimental Finance" (ABS 1), "Applied Economics Letters" (ABS 1), "Research in Economics" (ABS 1), and "Latin American Business Review" (ABS 1). His research has been awarded with a nomination for the Carolyn Dexter Award for Best International Paper (Academy of Management, 2024), as well as the Clovis L. Machado-da-Silva Award (Brazilian Academy of Management, 2023), two ENANPAD Best Division Paper Awards (Brazilian Academy of Management, 2016 in Accounting & 2023 in Strategy), a Best Plenary Section Paper (USP International Conference in Accounting, 2018) and was twice featured in SSRN's Top 10 Download List (2021 & 2024). Prof. Rodrigo is listed in the upper quartile of economic researchers in Brazil over the last decade, as measured by RePEc. He is a member of the Board of the Brazilian Finance Society since 2023.